

NBA Player PER Predictor

Regression Analysis Report

Objective

This project builds a linear regression model to predict a player's next season Player Efficiency Rating (PER) using their current season box score stats. Since official PER requires league-wide pace adjustments not readily available through the public API, a simplified proxy was used instead: a per-36-minute combination of points, rebounds, assists, steals, and blocks, minus turnovers.

Data

Player statistics were pulled directly from the NBA Stats API for the 2022-23 and 2023-24 seasons, covering 539 and 572 players respectively. Players with fewer than 500 total minutes were excluded to avoid noisy small-sample performance metrics, leaving 367 and 360 players in each season. Matching players across both seasons by name produced a final training set of 293 players. Eleven features were used: points, rebounds, assists, steals, blocks, turnovers, field goal percentage, three point percentage, free throw percentage, minutes played, and the current season PER proxy itself.

Model Performance

The data was split 80/20 into training (234 players) and test (59 players) sets, with features standardized before fitting a linear regression model.

Metric	Value	Interpretation
MAE	2.88 PER points	Average prediction error
RMSE	3.53 PER points	Penalizes larger misses more
R-squared	0.708	~71% of variance explained

Top Predictors

Ranked by absolute coefficient size after standardization, the strongest predictors of next season PER were: current season PER proxy (coefficient 4.38), minutes played (1.12), and rebounds (0.95). This makes intuitive sense: a player's current production level is the single strongest signal for their near-term future production, and playing time captures both coaching trust and durability.

Largest Prediction Errors

The model's five biggest misses on the test set highlight where it struggles most:

Player	Actual	Predicted	Type
Jrue Holiday	24.7	33.6	Over-predicted
Christian Wood	26.8	34.4	Over-predicted
Joel Embiid	54.0	47.1	Under-predicted
Kevin Love	35.8	29.1	Under-predicted
Day'Ron Sharpe	34.7	28.1	Under-predicted

Joel Embiid's under-prediction stands out most: the model predicted a PER proxy of 47.1 against an actual of 54.0, suggesting the model struggles to capture truly elite, outlier-level seasons. Jrue Holiday and Christian Wood were over-predicted, both plausibly tied to role or team changes between seasons that a single-season snapshot of stats cannot capture.

Key takeaway: A player's current production and playing time are the dominant predictors of next season performance, and the model explains roughly 71% of the variance in outcomes with an average error of under 3 PER points. It is most reliable for typical role players and least reliable for outlier, elite-tier seasons and players undergoing team or role changes.

Limitations to note: The training set is relatively small (293 players, 234 for training) given 11 input features, and results come from a single train/test split rather than cross-validation, so the reported metrics may shift somewhat under a different split. The PER proxy is a simplified custom formula and does not match the official PER calculation, which incorporates league pace and other adjustments.